# Trial of the Nine Realms

Supply Chain Disruption Impact Analyzer — an adversarial multi-agent system on the Runemark dataset

This document explains the problem, what each of the four agents does, the technology behind the system, and how each functional and non-functional requirement of the problem statement is met. Status notes say what is verified and how; nothing is claimed beyond what tests, saved logs or browser checks show.

## 1. The problem statement

**Context.** A ship is attacked, a port strikes, a government bans an export: one event can hurt industries in many countries, and experts disagree about how much. Most tools only summarise the news. Their output does not test itself, does not separate fact from opinion, and does not show how wide the risk range is, so decision makers cannot tell what is certain and what is disputed.

**The task.** Build a team of exactly four AI agents that treats every disruption as a court case. The user asks a question about a disruption, with an as-of date. The system then works in five steps:

- **Heimdall** turns the question into a Case File, gathers news and trade facts from Runemark and checks that the event is real.
- **Thor** argues, with evidence, that exposure is high.
- **Loki** argues, with evidence, that exposure is limited.
- **Thor and Loki** attack each other's claims in cross-examination; each attacked side defends with new evidence or concedes.
- **Odin** rules on every claim, sends the case back if the evidence is not enough, and writes the final briefing from his rulings only.

**The data: Runemark.** A fictional dataset written by the organisers: news articles, official notices, corrections, rumours and trade notes in PDF, TXT and DOCX, plus trade tables. It contains deliberate traps: documents published after the as-of date (a simulated clock), retracted and contradicted reports, duplicates, damaged and mis-encoded files, mis-tagged metadata and text that tries to give the AI instructions. Teams may not bring their own data; agents must find and read what they need themselves.

**What makes it hard.** There is no boss: Odin is a judge, not a planner. A single call that writes a risk report does not qualify; the agents must really search, disagree and change their minds. In the final round the organisers release *unseen* event packs and ask live questions, so anything hard-coded to the practice data fails. The output is one dashboard showing what is established, what is contested or conceded, and the low–high exposure range per country and sector, with every claim traceable to a Runemark document or trade-table row.

**Our worked example** (development pack E5): *“As of 25 January 2027, how exposed are Midgard and Tolan to the end of the Orvan corridor agreement? Give a low and high range for wheat and maize imports.”*

## 2. What each agent does

All four agents share one Case File and nothing else. Each has its own prompt, its own tool allow-list (enforced in code, not in the prompt) and its own private memory. Every model reply must match a typed schema; code then validates IDs, quotes, dates and numbers before anything is used.

### Trial flow (an explicit state machine; transitions are checked in code)

- **Run 1:** USER → HEIMDALL → THOR + LOKI (in parallel) → CROSS-EXAMINATION → ODIN → BRIEFING
- **Send-back:** ODIN → HEIMDALL RESEARCH → THOR + LOKI → CROSS-EXAMINATION → ODIN (run 2; never more than MAX_RUNS)
- **Early ends:** HEIMDALL → STOPPED if the event is disproved; any stage → FAILED with an honest limitation report

| Agent | Role | Tools (code-enforced) | What it produces | Rules enforced in code |
|---|---|---|---|---|
| **Heimdall**<br>Gatekeeper | Opens the case. Identifies countries, sectors, routes and event IDs; verifies the event as established / disproved / uncertain, checking corrections, retractions and contradictions; lists relevant trade tables and missing data. On a send-back, runs new targeted searches. | search, read_doc, get_trade_rows | The Case File (shared with all agents); in run 2, verified new evidence per missing item. | A status without a valid citation is downgraded to uncertain. Disproved stops the trial. Send-back queries must differ from every earlier query in the case. New evidence must be genuinely new (not seen before, not a duplicate). |
| **Thor**<br>The Alarm | Builds the strongest evidence-backed case that exposure is HIGH: the strongest plausible scenario, without assuming all trade is lost or that no substitute exists. | search, read_doc, get_trade_rows, trade_calc | Claims (fact or inference) with citations, assumptions and a low–high estimate per country × sector; attacks on Loki; defenses or concessions. | At least one search of its own per run. A claim with no valid citation never reaches Odin. Estimate bounds must equal results of its own calculation records. A defense must bring a citation not already used. |
| **Loki**<br>The Trickster | Builds the strongest evidence-backed case that exposure is LIMITED: substitution, other routes, domestic supply, inventories — only where documents show them. | search, read_doc, get_trade_rows, trade_calc | Same outputs as Thor, from the opposite side; searches independently and never copies Thor's claims. | Same checks as Thor. “Maybe it will be fine” is rejected: every claim needs a source and every number a calculation. |
| **Odin**<br>The Allfather (judge) | Rules on every claim, using code pre-checks (citations, dates, tiers, corrections, duplicates, damaged files, reproducible calculations) and the cross-examination record. Gives his own ranges, decides sufficiency, writes the briefing. | read_doc, trade_calc, validate_citation (no search: he judges, he does not investigate) | A ruling per claim (established / contested / conceded / rejected; fact vs inference), Odin's ranges, the send-back list of missing evidence, the executive summary and outlook. | Cannot rule “established” on a claim with hard failures; “conceded” only where the proponent conceded; ranges must come from calculation records, never averages. He names missing evidence but never assigns tasks. |

### Cross-examination, send-back and memory

- **Cross-examination:** each advocate must make at least one meaningful attack (a target claim plus counter-evidence or a concrete evidence gap). If one side fails, the briefing says the other side went unchallenged.
- **Send-back and run 2:** if Odin judges the evidence insufficient and names what is missing, code sends the case back. Heimdall runs new targeted searches; code checks the evidence is genuinely new; Thor and Loki revise, withdraw or add claims; Odin re-rules. No new evidence means an honestly labelled incomplete briefing. Runs are capped (default 2, maximum 3).
- **Private memory:** after each ruling, code writes general lessons for each agent (why a claim was rejected, which sources needed corroboration, which searches were useful), with names, IDs and numbers masked so no event answer carries over. Before planning, an agent recalls its own lessons, plans once without and once with them, and the log records the original action, the lesson used and the changed action (MEMORY_APPLIED).

# 3. Technology stack

| Layer | What we use | Why |
|---|---|---|
| Language and API | Python 3.12, FastAPI, Pydantic v2, Uvicorn | Typed contracts for every shared object (documents, trade rows, claims, attacks, rulings, briefing, log events); request validation. |
| Language models | Google Gemini (gemini-3.5-flash) via the google-genai SDK, with structured JSON output; the Anthropic backend stays selectable (LLM_PROVIDER) | Every reply is validated against its schema, with one repair attempt, bounded retries that honour the server's retry delay, request pacing, and token budgets. |
| Orchestration | Explicit state machine in Python (no planner agent) | Flow, limits and permissions are enforced in code and visible; illegal transitions raise errors. |
| Storage | SQLite: evidence store, append-only audit log (triggers block edits), private agent memory, calculation records, query cache; JSONL export | One file, reproducible; the audit log cannot be rewritten. |
| Ingestion | pypdf (PDF), python-docx (DOCX), segment-wise text decoding with charset-normalizer (TXT) | Handles mixed encodings, empty, truncated and damaged files without crashing; keeps hashes and flags. |
| Search | Hybrid: BM25 (rank-bm25) + one local vector index (fastembed BAAI/bge-small-en-v1.5, ONNX), fused with Reciprocal Rank Fusion (k = 60); metadata filters | No cloud dependency for search; country, sector, event, source type, reliability tier and date filters; the as-of gate on every path. |
| Calculations | Deterministic Python trade calculator (sum, mean, year-on-year, share, offsets) with replayable records | The models never do arithmetic in their heads; every number links to its input rows. |
| Dashboard | React 18, Vite 6, TypeScript, Tailwind CSS 4, Server-Sent Events (polling fallback) | Live view of real recorded events only; click-through to documents and trade rows. |
| Quality | pytest (336 tests, scripted model for isolated tests); Microsoft Edge via playwright-core for browser checks | Security, failure and limit behaviour tested without network calls; UI checked against API data. |

## Architecture in one picture

| | |
|---|---|
| **Dashboard** (React) | question panel · evidence feed · trial view · exposure · briefing · trace |
| ↓ | HTTP + Server-Sent Events |
| **FastAPI** | POST /cases · GET /cases/{id} · /events (SSE) · /docs · /trade · POST /ingest · /feed |
| ↓ | |
| **Trial state machine** | flow, run limit, cross-exam rounds, budgets, send-back decision, briefing |
| ↓ | |
| **Agents** | Heimdall · Thor · Loki · Odin — own prompt, own tool allow-list, own private memory |
| ↓ | |
| **Services** | LLM client (schema validation, one repair, retries, pacing, token budgets) · hybrid retrieval (BM25 + vectors, RRF, filters, as-of gate) · trade calculator · citation validator |
| ↓ | |
| **SQLite** | evidence store · append-only audit log · agent memory · calculation records · query cache |
| ↑ | |
| **Ingestion** | PDF / TXT / DOCX parsers, trade tables, damage and injection flags, duplicates and corrections |

# 4. How the requirements are met

Status key: **Done** = implemented and verified by tests and/or a real run; **Pending** = implemented and tested, real-model demonstration still to run.

## 4.1 Functional requirements

| ID | Requirement | How we meet it | Status |
|---|---|---|---|
| FR1.1 | Question → Case File with as-of date | Heimdall's Case File: countries, sectors, related countries, routes, event IDs, verification, trade tables, missing data; the as-of date is fixed by the user and cannot be changed by any agent. | Done |
| FR1.2 | Gather documents and trade data, any format | Ingestion of PDF/TXT/DOCX, manifests in mixed formats, prose and fixed-layout trade tables; live packs via POST /ingest (folder or validated zip). | Done |
| FR1.3 | Verify the event; retractions, contradictions | Established / disproved / uncertain with citations; corrections and retractions linked at ingestion; a disproved event stops the trial and the user is told why. | Done |
| FR1.4 | Re-search on send-back | Per-item new targeted searches; repeated queries refused in code; novelty check on new evidence. | Done (tests); real run Pending |
| FR2.1–2.3 | Thor: high case, evidence, defence | Own searches; claims with doc IDs and calculated estimates per country × sector; defends with new citations or concedes. | Done |
| FR3.1–3.3 | Loki: limited case, evidence, defence | Same mechanism from the opposite side; substitutes only where documented. | Done |
| FR4.1–4.2 | Odin: rule every claim, fact vs inference; classify | One verdict enum everywhere; unruled claims reported; code guards on established / conceded. | Done |
| FR4.3 | Sufficiency check and send-back | Structured missing-evidence list; code decides the send-back within the run limit. | Done (tests); real run Pending |
| FR4.4 | Briefing from rulings only | Built from Odin's rulings; every citation and calculation re-checked at build time; failures dropped and listed. | Done |
| FR5.1–5.3 | Flow; parallel claims; at least one attack each way | Explicit state machine; Thor and Loki in parallel threads; missing attacks reported as unchallenged. | Done |
| FR5.4 | At least 2 runs with new evidence; hard limits | MAX_RUNS (default 2, max 3); new evidence verified; no new evidence → incomplete briefing; LIMIT_HIT logged. | Done (tests); real run Pending |
| FR6.1 | Private, lasting memory per agent | SQLite memory bound to one agent in code; persists across cases; only the Case File is shared. | Done |
| FR7.1–7.4 | Dashboard: input, live feed, trial view, exposure, click-through | React dashboard: question panel, evidence feed with badges, Thor vs Loki with rulings, range bars, citation → highlighted paragraph or row. | Done |
| FR8.1–8.3 | Briefing: findings, exposure + outlook, citations | Executive summary; established / contested / conceded (+ rejected); Thor / Loki / Odin table; outlook with assumptions; inline [doc, paragraph or row, date]; source list; Markdown + JSON. | Done |

## 4.2 Non-functional requirements

| NFR | How we meet it | Status |
|---|---|---|
| NFR1 Agent architecture | Four agents, each with its own prompt, tool allow-list and memory; no central planner; Odin judges but never assigns work; the flow is a state machine, not a two-agent loop. | Done |
| NFR2 Grounding | No claim without a valid citation (rejected in code otherwise); quotes must match the stored text; fact and inference tagged on every claim and ruling; every citation resolves via /docs or /trade (tested); all sources are Runemark and labelled as such; missing evidence is stated, never filled in. | Done |
| NFR3 Reliability | Empty, truncated, mis-encoded and damaged files handled with flags; failed tool calls and empty results reported to the agent; bounded retries then a controlled error; one advocate failing → “unchallenged”; never “complete” when evidence is insufficient (incomplete / partial / failed labels). | Done |
| NFR4 Observability | Append-only audit log of every query, retrieved item, claim, attack, defense, concession, ruling, memory read/write/applied, injection attempt, clock block, limit hit and error; live on the dashboard; JSONL export. | Done |

| NFR | How we meet it | Status |
|---|---|---|
| NFR5 Performance | Hard limits on runs, cross-exam rounds, tool calls and tokens per agent and per case, retrieved evidence, retries, request time and upload size; identical queries served from a logged cache; send-back queries must be new. | Done |
| NFR6 Reproducibility | Per run: Case File, as-of date, claims, cross-examination, rulings, send-back, new evidence, configuration and audit log saved; calculation records replay exactly; the briefing traces to document paragraphs and trade rows. | Done |

### 4.3 Security (prompt injection and the simulated clock)

- Retrieved text is wrapped and escaped as untrusted evidence; every prompt says it cannot change roles, tools, schemas or limits. Detected instruction text is withheld from agents, kept in the original and logged (TRAP_INJECTION).
- The as-of date is enforced in code on every document, trade-row and calculation path; later documents are blocked and logged (CLOCK_VIOLATION_BLOCKED). Tests show injected text cannot skip cross-examination, change the question or force Odin to approve a claim.
- Nothing from the practice pack is hard-coded: the same pipeline ingests unseen packs (tested with a live-added pack).

### 4.4 Verification and open items

- 336 automated tests pass. A full real case (Phase 3) ran end to end: 11 claims, 6 attacks, 2 defenses, 4 concessions, 11/11 ruled, briefing with Thor / Loki / Odin ranges; saved in logs/demo/.
- With Gemini, real cases ran through the dashboard live up to cross-examination (9 claims, 3 attacks) and logged a real memory-driven plan change (MEMORY_APPLIED).
- **Pending:** a complete two-run real case (send-back plus new evidence in run 2). It needs more model quota than the free Gemini tier gives per day (about 25–30 requests per model per day versus roughly 60–80 per case): enable billing on the AI Studio project, or use the Anthropic key after 1 November 2026.